

What's the Look of "Negative Gender" and "Max Ethnicity" in AI-Generated Images? A Critical Visual Analysis of the Intersectional Politics of Portrayal

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ABSTRACT

In this exploratory paper, we focus on intersecting design political and visual processes of gendering and racializing in online AI image generators, in particular ArtBreeder and Midjourney. While AI image generators are becoming an integrated part of our contemporary society, they draw on cultural and historical imaging conventions of sorting and ordering the world and the people in it. These tools' powerful visual rhetoric can potentially aggravate existing discrimination, if not critically reflected upon. We argue that these design-facilitated representations position the 'user' into cultural imagery of representations with political implications. With an intersectional perspective from the feminist visual analysis, we critique and uncover how gender and ethnicity are represented and built into the systems, both in terms of visual culture and in designed interactions. We problematize these design strategies, and urge the HCI community to engage in further design political inquiries regarding the visual culture mediated by AI image generators.

KEYWORDS

Generative AI; AI Art; Critical AI, Intersectional AI

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1 INTRODUCTION

While visual culture has been central to human societies since their early beginning [5], recent years have seen a sharp rise in a very specific form of producing and creating visual culture: AI-enabled visual culture. While various tools from printing press to photography, have been part of the production of visual culture throughout history, we are now entering a time in which AI technologies are used for creative purposes in production of everyday media. The use of AI technologies for visual creation, including applications

such as Midjourney¹, ArtBreeder², and StableDiffusion³, and the embeddence of AI features to well-established creativity support tools (CSTs) such as Photoshop and NVIDIA Canvas⁴ are becoming increasingly common. A large amount of the increasing popularity of AI technologies for visual generation can be attributed to the recent emergence of Diffusion Models (DMs) [29], which has improved the generative capabilities of AI systems - to an extent. As these visual generative AI (VGAI) tools are adapted to use in the socio-technical fabric of the society, they are posing various ethical and sustainability questions to the societies adopting those very technologies to use [6, 10, 21, 22, 30].

When the creation of aesthetics and visual culture is automated with the emergence of visual generative AI, there emerges a need for *critical visual cultural inquiries* of the imagery produced by AI. Historically, such visual cultural analysis has been conducted in fields such as visual design, art history, gender studies, feminist science and technology studies (STS), and we argue that we need to draw from these disciplines and knowledge traditions in order to develop methods for a critical analysis of aesthetics and visual culture generated by visual generative AI. Furthermore, we argue that designers of visual generative AI (VGAI) tools have the responsibility to think not only about biases built into the models, but to acknowledge the design political positionality in creating these VGAI mediated visual cultures. We argue, that there is a need to develop critical methods for proactively addressing the questions on automatizing visual culture in (design) practice, prior to adopting the VGAI systems to use in the society. These efforts should be particularly focused on uncovering how the imagery constructs privilege and discrimination of certain people, from the perspective of design justice [11] in regards to social categories, and intersectional perspectives are often at the core of exploring such questions.

In this exploratory paper, we critically explore these under-explored dimensions of cultural politics of visual representation in the design of these VGAI systems. This politics of representation refers to the negotiation over the meaning of images and conventions of portrayals in society and culture. They are embedded, in and generative to, power relations and structures, which can be surfaced by analysis [4, 15, 34, 36]. Such analysis, combining insights from the interdisciplinary fields of feminist and anti-racist

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¹<https://midjourney.com>

²<https://www.artbreeder.com>

³<https://github.com/CompVis/stable-diffusion>

⁴<https://www.nvidia.com/en-us/studio/canvas/>

scholarship and visual culture, propose an ethical, if not down-right political, stance towards cultural experience and technological representations thereof; insisting on a broader approach for the representation of humans to accommodate the intersectionality of gendered and racialized subjectivity, including aspects of class, age, ethnic, national and sexual identification, without privileging a default whiteness or universal masculinity as the sign for all humankind. The cultural technologies of gender, race and empire drive much of the present assumptions about human identity, now and in the past. Everyday algorithms reproduce and multiply our cultural biases on a global scale by way of media and design [39]. There is a distinct risk of increased polarization in society with simplified categorizations of groups of people, unless we become more cognizant of the politics of representation in an age of rampant individual exceptionalism and technological enhancement of culture. In this paper, we contribute to this discussion from an intersectional perspective, focusing on gendered and racialized visual representations in generative AI systems, and their relation to each other. This analytical approach is borrowed from feminist and anti-racist cultural research [12, 15]. In practice, we set out to conduct a first-person visual exploration from an intersectional lens. We focused specifically on ArtBreeder and Midjourney as a few of the popular internet-based VGAI tools. We will walk the reader through illustrative examples of the design of these systems, and highlight design and development decisions that arguably contribute towards certain kinds of practices in racialization and gendering. We problematize these design choices and representations created by them while providing historical and cultural contextualization to demonstrate the ethical questions in a more concrete way for the reader.

Finally, we suggest concrete considerations for designers and developers as strategies to avoid the problems found within these systems, and highlight the need of further research in how designers could address their role in cultural politics of visual representations in the context of Generative AI systems. We also highlight, how there is a need to bridge the gap between visual cultural analysis and generative AI technology, and to frame the generative AI imagery in the holistic context of the prevailing visual culture and imaging conventions within it.

2 BACKGROUND

2.1 Design Practices that Lead to Genderizing and Racializing in VGAI Technology

We start by briefly covering genderizing and racializing in the context of VGAI systems, while putting emphasis on the design practices and cultures that lead into such characteristics becoming embedded into the AI technology. While gendering and racializing practices in the design of AI technology are acknowledged and studied to some extent, they are still present in many technologies from voice interfaces [13] to machine vision [6, 30], robotics [31], and generative AI. In regards to gender, racialization, and AI, prior research by Scheuerman et al. [30] has looked into how gender biases are built into the systems that utilize Machine Vision, and Sender et al. [31] and Sparrow et al. [33] have problematized the design of characteristics of ethnicity in the context of robotics. Similarly, Buolamwini et al. [6] have discussed related problems within facial recognition software, and Buolamwini [7] has also discussed

racializing biases in AI. Taking this discussion towards the design practices that enable embedding such problematic characteristics in the systems, Dray et al. [14] point out, how gendering is often not actively considered in design processes, but rather embedded sub-consciously by the designers of the systems. They argue, that the way gender is constructed in design processes and its effect on the designed technology artifacts is not well understood [14]. Feminist HCI has aimed to set a design research agenda for addressing these questions in the design of technology [3]. However, it remains still unexplored territory when it comes to the newly emerged VGAI technologies.

Feminist HCI argues for considerations of alternative viewpoints, agency, and developing reflexive methods to understand viewpoints of people [3] and according to the legacies of feminist research, reflections on implications for their sense of self-hood (subjectivity). Bardzell's call for reflexivity in feminist HCI motivates our exploratory study, which argues for the need of methods in which designers could reflect on and consider the politics of portrayal in the context of designed VGAI technology – prior to these technologies being deployed to use in the world. *It is our aim with this paper to specifically engage in and develop reflexive knowledge on how processes of racialization and gendering are embedded in VGAI.* However, if we are to set a serious agenda in addressing these design 'injustices' already at the stages of design and development of the generative AI technology, we need to understand better how these processes of gendering and racialization take place in the socio-technical assemblages of VGAI. This involves uncovering how gendering and racialization take place in techno-material practice and situated design and development processes. In this empirical work, we begin by unpacking the biases that are specifically embedded in materiality and artifacts of the VGAI technology.

2.2 Visual Politics of Portrayal and Constructing Power Structures

Visual cultural and art historical inquiries on aesthetics and visual politics of portrayal have been traditionally seen as something outside of the scope of HCI inquiries. However, with automatizing the generation of visual culture, there is a pressing need to bridge this disciplinary gap and acknowledge the visual products of these tools in the context of the wider visual culture. Considering the visual culture in a wider sense also opens up questions of power structures, and how they are configured through visual representations. For such open-ended insights we need the intersecting perspectives of anti-racist feminist theory and cultural studies of science and technology in society, or in short, feminist STS [40]. In this view, is important to look at the gendering and racialization practices in design of technology as a process of constructing power structures by way of categorizations in the society through visual culture and aesthetics. Through setting certain norms and values (valuations of whom gets to matter, who is represented and integrally part of the technologies), power structures and inequalities that are part of society are reinforced and created. For example, when it comes to gendering, very often, we can also see the visuals produced by VGAI systems highlighting the heteronormativity of femininity in what visual theorist Laura Mulvey called their "to-be-looked-at-ness" [27] - a spectacular "doing of femininity" through make-up

and attractiveness (to men), artificiality, superficiality, vanity, and pretense, that has produced a derogatory categorization of women. But how did these representations end up in the VGAI systems? The conventions in wider visual culture (training images of the AI) as well as what kind of imagery the engineers and designers of the VGAI systems have been exposed to and that have formed their perspective of the world have all played a role in embedding these harmful categorizations and norms in the VGAI systems. Thus, we argue that the problem should not be narrowed in the technological artifact, but rather we should frame the problem in a wider sense in terms of visual culture as generative and regenerative in itself of power structures and identities [4, 12, 18, 20, 36].

Historically, these mechanisms of the visual politics of portrayal have been used effectively for constructing hierarchies of power and discriminating certain groups of people. These norm constructed in visual culture can be used as a basis of oppression, as they are exclusive to people who do not fit into the norm, and they communicate how an individual 'is ought to be', and to elevate the social standing of a certain group of people. For example, constructing a visual norm of an ethnicity that is very narrowly defined is a derogatory visual rhetoric strategy that the Nazi's used in an attempt of setting a 'visual archetype' of an ideal person (and idea of a "us" and "them"). Similarly, we see how such idealized norms are set in visual social media platforms, such as Instagram, that distribute visual imagery. There are many other similar visual cultural examples in the history of visual culture – and particularly that of Euro-Western visual culture. For example, another case of constructing social arrangement based on visual features is the 19th century science of physiognomy. It centered around making up facial ideals based on the idea that one can judge the character of another person from their facial or bodily characteristics, and emerged historically in Europe alongside the visual technology of photography and the racial sciences of eugenics (racial hygiene) and phrenology [1, 8, 26, 34]. While the science of this is long debunked, and the cultural legacy of it critiqued in scholarship [4, 26], the physiognomy fantasy still prevails in present day social media and the visual cultures of popularized science [9, 37, 38]. While these examples, of course, are radical and were also often deliberately used to drive a racializing agenda, we would like to make a parallel in our paper to how same visualization approaches of sorting, ordering, and constructing harmful normative representations are deeply built into the the AI systems that – very similarly – arrange facial visual features based on their proximity to other features, and their color values, such as skin tones. In these statistical AI models that analyze images based on their pixel content and relation, we can easily make generalizations that end up harming certain groups of people – which we will demonstrate and discuss throughout this paper. This is something that other scholars have discussed as the "coded gaze" [7]. Thus, we argue that there is a need to understand and address the involved problems that relate to such practices of arrangement and sorting of visual features and characteristics, particularly when it comes to gender, ethnicity, and power, in the design of VGAI systems.

3 METHODOLOGY

There is largely an absence in *formalized methodologies* to assess the cultural, gender, and ethnicity based biases in generative AI tools, as well as conducting intersectional or visual cultural analyses for their visual imagery. There is preliminary work conducted in the domain (e.g. [2, 24, 28]), but there is yet a need to find formal methods for such analysis. Developing approaches for such formalized analysis, however would be essential in being able to bring a more critical outlook in the design phases of such technology, particularly in a setting in which ML engineers and other members of the design teams can not be expected (are traditionally not trained) to have deep competency of qualitative and critical visual cultural analysis. This paper is also contributing through an exploratory approach to the development of these methods. This called for a rigorous documentation of the exploratory approach that we used in the studies.

We draw from prior media studies in terms of the theoretical positioning that informed our qualitative analysis, as well as in how we used *critical visual analysis* to unpack the visual politics of portrayals in the AI generated imagery. Furthermore, we extended this approach towards *critical design analysis* at times, when the interactions and interfaces were examined, outside of the scope of the visual imagery itself. While critical visual analysis is an interdisciplinary method that has been used in attempts to make sense and contextualize images, there are many ways to conduct such analysis as the method is not strictly formalized [17]. Often, however, the critical visual analysis is interpretative and includes elements of description and documentation, reflection, and analysis [17] with the aim to uncover meaning. According to Sturken and Cartwright [34], this meaning takes place in three dimensions; in viewers interpretation, the meaning embedded in the thing being analysed, and the context in which it is seen. Critical visual analysis has been used in HCI research in very limited occasions (*see* [32]). Then, through critical analysis, each of these sites and their relation is analysed. While in our analysis, the viewers interpretation is implicit, the meaning in the imagery artifacts and their societal context is a core part of our analysis, as well as their relation. Critical visual analysis relies heavily on the subject expertise of the persons conducting the analysis.

In our case, the qualitative analysis of the imagery and designs was conducted by reviewing the material in an iterative manner in relation to the media and culture theories summarized in Section 2, by both of the two authors. While the first author has a formal international (Nordic/SEA/SA⁵) education in HCI and design, visual communication, and experience in conducting qualitative visual analyses, the second author is a Nordic expert in feminist cultural studies theories, STS, visual culture and interdisciplinary more-than-human research. Such fields, when brought together as feminist forms of posthumanities, forge possibilities to pursue experiments with new forms of knowing and creating with, and caring for, how technologies in society – like VGAI – to stabilize or destabilize cultural conceptions of humanness and humanity. Our analysis should be seen in this light, as an experimental attempt in admixing visual knowledges and visual practices, and coming from such feminist, queer and anti-racist calls. These calls urge us

⁵Nordic = North Europe, SEA = South East Asia, SA = South Asia

to decolonize knowledge production, to decenter human exceptionalism, and to practice care and concern for how technologies both make and unmake us. Drawing on such scholarship, both authors embrace critical epistemological viewpoints of how research always is limited, partial, embedded, embodied and situated knowledge in practice [19], which also carries on to the performed critical visual and design analysis. In performing the analysis, the authors were keen to recognize the stark power differences and legacies inherent in Euromodern imaging conventions of representing gender and ethnicity in particular. For the sake of delimiting ourselves, we chose to start out with gender and ethnicity and striking highlights of their intersection in the Euromodern archive and social imagination. While this focus on sexist and racist visual traditions risks reiterating the problem we want to obliterate with our critique, the air of white innocence around new VGAI forces us to revisit its Euromodern legacies, image conventions and visual traditions, however disturbing they may be. Another limitation to the focus on gender and ethnicity in our analysis is that it leaves out many other possible intersectional dimensions of the VGAI's politics of representation, such as ableism, ageism, heteronormativity, class-norms, speciesism, etc. Some of these intersections are part of our extended work outside the scope of this publication.

In Figure 1, we have specified the approach of taking visual samples out of both of the systems; ArtBreeder and Midjourney. We often took samples of 20 in each of the parameters (gender, ethnicity, different prompts), while in the case of wanting to see how the system would randomly generate images of faces, we took a larger qualitative sample of 100 images. We extracted these samples, and performed a visual cultural analysis for them.

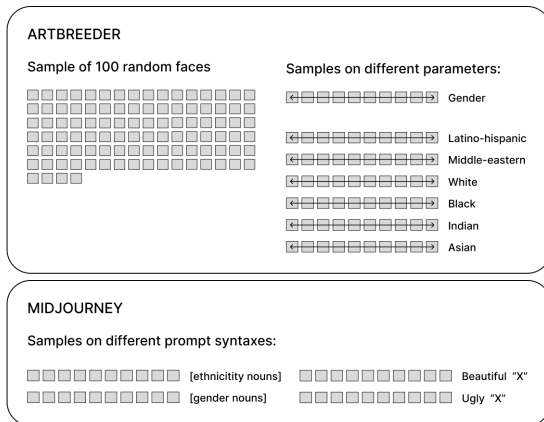


Figure 1: Approach for taking visual samples of generative AI tools

4 RESULTS AND CRITICAL ANALYSIS

4.1 Problematising Racialized and Gendered “Archetypes” and “Antitypes” as an Approach for Constructing Visual Polarization and Hierarchy

In the first exploration, we focused on the statistical representations of ethnicity and gender in the generative AI application ArtBreeder. Effectively, what we could see in the system was practices of arranging people in categories and in relation to each other based on their visual features (such as skin color, eye color, hair length, facial shape) to construct social hierarchy. These types of categorizations/arrangements create what we propose to refer to as in the context of designing the systems as visual norm-producing archetypes and antitypes. In these archetypes, the maximum/core representation of a gender or ethnicity is defined, and in the antitypes the opposite of that is defined. This establishes both norms and polar opposites of those norms, defining simultaneously both what ‘a caucasian person’ is ought to look like and also what it is not supposed to look like. We propose the term of *Archetype* and *Antitype* as concrete ways of discussing the polarised representations in the context of VGAI systems, as it allows us to refer to them in the subsequent parts of this paper more conveniently. These terms can also be seen as our attempt of concretizing the specific problems that are part of training AI models. After developing these terms during the empirical work, we also saw that the term of archetype has been used prior in media studies [25] for cultural vehicles conveying messages. In Figure 2, we showcase both the gender and ethnicity based archetypes and antitypes (the maximum and minimum representations). While both of the systems produced these gendered and racialized representations, Artbreeder had also interface elements that very concretely visually supported such categorizations. In Figure 3, we can see the slider designs for determining the “minimum and maximum race”, as well as polar opposites of the gender.

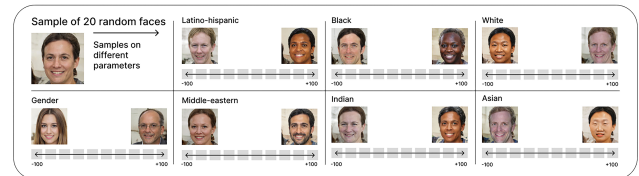


Figure 2: Artbreeder slider designs for determining the minimum and maximum for race and gender

In the case of Artbreeder, images sampled from these archetypes and antitypes are produced by the user by controlling a slider, seen in Figure 3. Using this slider, the user is able to control the parameters to produce the max (100) or min (-100) representations of a certain racialization and gender. In Figure ?? we present archetypes produced by ramping the values to the maximum in each category, which we did using 20 different seed images. By repeating the process with several images, we can see the repeating patterns and common features emerge in the images. It clearly is problematic in the first place, that such ethnicity-based labels like “asian” or

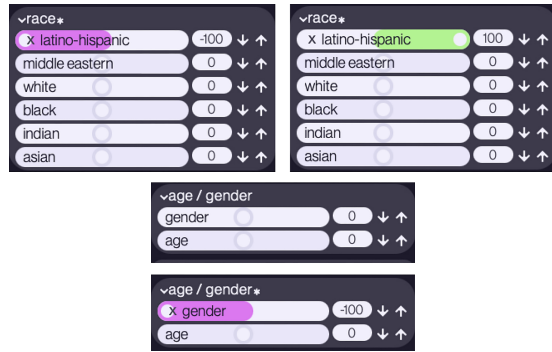


Figure 3: Racializing and gendering slider designs in ArtBreeder

"white" have been designed into the system interface in the first place, but even more so, when we start to see how visual characteristics are coupled with these labels. When we looked at the archetypes ("max" in Figure 2) and antitypes ("min"), we started to also see some patterns in polarizations between them. The features of the "white" archetype are most often applied as the polar opposite of what the system categorizes as asian, hispanic, indian, or black. Thus, we can see these social categories being juxtaposed in relation to each other. Essentially, the system is producing arbitrary polarizations between certain representations of people based on their visual features. It feels absurd to think, that one could arrange humans in such a relation to each other based on facial features. Yet it follows the historical patterns discussed earlier in Section 2.2. We raise questions regarding what are the concerns in doing this kind of seemingly arbitrary "sorting of people" based on gender and ethnicity in VGen-enabled visual culture. We can anticipate that facial archotyping and antityping are problematic for several reasons. For instance, they are; *constructing a visual norm of a race and gender that is very narrowly defined, highlighting the differences and constructing a hierarchy between racialized and gendered representations, and pairing certain arbitrary aesthetics with certain racialized and gendered representations*. At an overarching level, these factors contribute to new colonialization through visual culture and serve the agendas of certain world-views by enforcing certain power structures.

4.2 Constructing Binary Gender Norms and Connecting Aesthetic Judgements of Beauty with These Gender Norms

When we looked into the gendering processes in ArtBreeder, we noticed that there is automatic gender-assigning by the system in connection to racialized archetype, and that the system is constructing heteronormative portrayals of gender in which the feminine portrayals are often sexualised. Thus, the main concerns in regards to gender are; *enforcing heteronormative gender norms and assigning gender to racialization automatically and arbitrarily*. For example, the "max-asian" is often feminized, whereas the max-white and max middle eastern are more often masculinized, for example, by adding a dark beard. Furthermore, we can see that most often the "commonly racialized" groups such as indian, asian, latino-hispanic,

and black seem to have feminine facial characteristics. These visual representations seem to align with feminist critiques of the universal man (representation of a white man), and how racialized differences from that norm entail a disempowering "feminization" [20]. In regards to parameterizing gender, the ArtBreeder tool incorporates a feature that we called a "binary gender slider" (see Fig. 3). This design feature is controversial in many ways, as it enforces the concept of binary gender and as a consequence is not inclusive to gender minorities. It is also actively gendering representations of people. The "binary gender slider", similar to the racializing slider, has -100 (feminine archetype) and +100 (masculine archetype). The binary and heteronormative design for the slider conceals many problems: *producing imagery that follows and sets a certain norm of binary gender (e.g. female has large eyes and lips, or make-up applied, the male is bald or has short hair, arranges the parameter space to two polarizations rather than including diverse representations, and trivializes and hides the complexity of gender by making it a simple matter of sliding a button*. Furthermore, it is impossible to produce gender norm-agnostic representations, where one would mix and match various feminized and masculinized features together in a desired way. Rather, these features are coupled in binary gender norm adhering arrangements. While these same patterns are also visible in the prompt-based system Midjourney, in those one has to also specifically prompt for "female" or "male" to produce these norm-adhering representations. In regards to the gender slider, the system also "broke" when we input an image that already corresponded to a gendered stereotype of a feminine face, and turned the feminine parameter to maximum (or minimum, in this case). The feminine characteristics enlarged the eyes and lips of the face until a point, in which the eyes and lips became unnaturally large. In fact, the very maximum of femininity seems to imply a certain conception of "beautiffulness" as we came to discover when we used this term as prompt for Midjourney-imagery.

As beauty norms were clearly infused in the feminine representations produced by the system, in a subsequent exploration we explored Midjourney in terms of what it considers as beautiful and what it considers ugly. In the examples provided by Midjourney, we can see that the prompt of beauty involves often women, and particularly young, heteronormative representations of women (see Figure 4 and 5). On the right side, we can see what prompts 'beautiful passport photo' and 'ugly passport photo' produce. The "ugly" passport photo seems to have heteronormative masculine visual features, whereas the "beautiful" photo contains a seemingly young white heteronormative female visuality. The same pattern was reproduced in other completely absurd contexts as well. For example, we input various things in the syntax "beautiful X", replacing the X with hat, coffee, etc. – various random things. In Figure 4 we can see, how the application is trying to insert a face of a young woman into the coffee cup, as it has been trained to value that as beautiful. This is no doubt an example of a harmful norm-construction, as it is portraying specific features in various abstract contexts as the norm of beauty. Furthermore, all of the human portrayals of beauty and ugly represent a "white" face, as it would be categorized in Artbreeder. Questions can be raised regarding if these are really the kind of visual culture that we want to facilitate and construct by designing these systems. Furthermore, questions are raised about who or whose labels is the system trained on. In this case, unfortunately,

it seems that the labeling practices have also placed female as the object of spectatorship, aligning with priorly criticised observations of media culture, while elevating "whiteness" as the norm.



Figure 4: Prompts "ugly passport photo", "beautiful passport photo" and "beautiful hat" produce gendering imagery in relation to aesthetic judgement

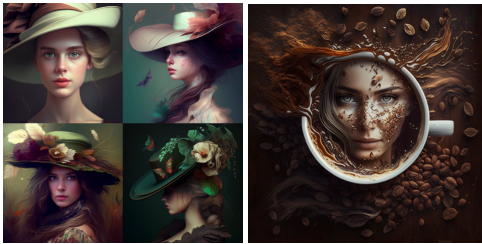


Figure 5: Female face inserted into coffee cup as a result of prompt "beautiful coffee" and into image of a "beautiful hat".

4.3 Constructing Norms of Social Hierarchy for the Gendered and Racialized Representations

The question is not only concerning gender and individuals' beauty, but also constructing social norms in how those genders are juxtaposed in relation to each other. An example of this are the images in Figure 6, in which were created with prompts "beautiful family" and "ugly family". In both cases, we can see that the family is white and represents a nuclear family with (mostly) one or two parents and one or two children. These images seem to communicate that a white nuclear family with traditional binary gender aesthetics (and roles?) is the norm. And, if we analyze the images further, there is an uncanny resemblance to the Nazi propaganda posters presenting images of an ideal family (see also in 6). In these propaganda posters and images, a caucasian, happy nuclear family was represented in an order to create an ideal for people to strive towards. It is extremely concerning to see, that similar ethnocentric representations are automatically produced by generative AI tools.

Finally, we looked at representation on a large scale. When requesting to produce random images, the Artbreeder system generated mostly caucasian-looking adult faces (see last image in Figure 7). There were also a few asian and children as samples (highlighted), but we can see here a complete lack of diversity in terms of what the "random feature" produces. Most of the generated faces are



Figure 6: Constructing norms of social hierarchy in family representations in Midjourney, family aesthetic similar to Nazi posters [35]



Figure 7: Large sample on randomly generated images with ArtBreeder that mainly are white

from a certain age and racial representative classes. It also seems that feminine faces are around 35 of the representation. While this may be partially a result of what the users have fed into the system (the system may use user-fed images as training material for their system), the same patterns and problems were visible in the prompt-based system Midjourney, in how we saw that it produced white faces as a default.

5 DISCUSSION

As discussed in Section 2.2, there is a long history of racialized and gendered "categorization and sorting", which is also visible in modern day generative AI technologies. These historical contexts of racialization can be paralleled with the Artbreeder and Midjourney system designs, in which specific facial features are coupled with racialized representations. Constructing racialized representations that are arranged in relation to each other in ways that highlight differences and construct a hierarchy (the fact that a white person is always the opposite of latino-hispanic, asian, or black) is a concrete design and development choice of "coding gaze" [7] that builds visual polarization and enforces certain visual legacies of power differences, in a modus of aesthetic choice. This categorization and sorting is also seen in how gender is arranged within the systems, often in a manner that enforced binary gender conceptions – by coupling visual features with binary mappings and representations.

5.1 Methodological Reflections

While the nature of the tools influences in what kind of agency the user has in generating the images; ArtBreeder allows "random generation" and has restricted the user into certain parameters, such as "gender" and "race" (as defined by the system designers

in the interface), Midjourney is prompt-based. That influences on the way the analysis is conducted, and puts also more emphasis on the interface design choices of the Artbreeder tool, whereas with Midjourney the emphasis seems to lie of the training data and how that was labeled, and fed into the model. This is to say, that *different generative AI tools require different approaches for analysis* in further development of such critical analytical methods for investigating these tools and the imaginaries they create. This also argues for an empirical approach, developing the knowledge ground through these two empirical cases. There is plenty of work required to investigate the various different generative AI tools, such as DALL-E, Stable Diffusion, Adobe Firefly, etc. – as they all have been trained in different (yet often similar) ways and with different training data, and practices in labeling and design. Furthermore, we need more empirical knowledge of the design processes of these models – what kind of decisions are done in those processes, by whom, and why. Taylor et al. [23] have provided initial efforts to do such analysis of data labeling practices, for example, and we need to extend this type of research to the developing VGAI technologies - designers and developers that actively construct them. After performing a series of explorations with ArtBreeder and Midjourney, many questions are raised. For example, have some of the problematic design features covered in this paper surfaced in the design processes of these systems? Have they been ignored? Or have all been embedded unconsciously and these questions have not even come forth in the process? Who is labeling and gazing whom? By answering these questions through empirical studies of the design and development processes of these systems, we can address further how and why these biases are built into the systems in practice, and develop methods of avoiding or encouraging certain types of gendering and racializing practices in the design of VGAI.

Many methodological questions emerged throughout the process regarding the methodology; how many samples would be enough to show representative patterns? In a systematic approach to study visual imagery of the tools from an intersectional lense, what prompts and syntaxes are centrally relevant, and which ones less? Why? While many of these questions are difficult to answer, we engaged in an exploratory process in which one route of inquiry left to identifying the next one; for example the gender imagery revealed the connection to aesthetics of beauty, and through this process the analysis started unfolding. Potentially, through empirical ground up research, we can start developing knowledge that will help to methodize and systematize this type of analysis, so that it would be easier to conduct within the design processes of VGAI technologies. But furthermore, we have to be mindful to the consideration that such analysis relies heavily on the subject matter expertise of the persons performing the analysis, and to think how this expertise could be brought into the design processes of VGAI early on.

5.2 The Urgency of Problematizing Embedding Aesthetic Judgements With AI

In this paper, we have raised questions and problematized, what happens when certain aesthetic judgements are built into the systems - and particularly when these imagery and tool architectures are applied to common use in media. Tools are already being developed to utilize these technologies in image production for movies,

visual design, and other disciplines that could benefit from such generative capabilities. But in applying the imagery produced by these tools to everyday media productions, there is an even larger danger in effectively enforcing racial, gender, and other harmful stereotypes and norms that can exclude and alienate groups of people. Thus, it becomes extremely important to find strategies to avoid gendering (and racialization as discussed earlier) in visual cultures constructed through the use of these tools. It is crucial to avoid 'mass production' of certain types of aesthetics that set beauty norms that perform such racializing and gendering categorizations. We need to seriously reflect on the visual politics of portrayal by raising questions about who is gazing whom, and this also urges us to position ourselves design politically [16]. When the data used in ML models is labeled, it can become a gendering and racializing practice, in which some people are gender-assigning and race-assigning others based on visual features and norms. This puts the persons having agency in labeling as the 'gazers' and the people represented in the training imagery as the 'objects of the gaze'. As discussed earlier in 2.2, this gaze or act of looking has been described in earlier research the complex power relations that are constructed in the acts of looking and being looked at. In early media studies, the gaze of the spectator was identified as implicitly male, objectifying and "othering" (not-self) the women on the imagery [20]. Contemporary feminist scholars have complicated the original gendering of spectator and visual object, and now discuss different kinds of gazes distinguished by sex, gender, race, and class, and these gazes can be deployed by different kinds of spectators [4, 20]. What is surprising, is that these long-standing critiques of visual culture are nearly absent in the development of such systems that effectively produce visual culture, powered by AI.

5.3 'Unlearning' the Generative-AI Systems

As mentioned above, we can build on what is previously known in research on gendering, racialization, and the representation of human faces in visual culture. The "cultural archive" of representing humanness that is shaped by certain norms is a process that carries on and 'leaks over' from other parts of the visual culture to the design of generative AI systems. - and vice versa. While our brains are wired to think in terms of pictures and categorizations - we can unlearn and re-learn these categorizations. Thus, we argue that VGAI systems need to unlearn and relearn certain associations that relate to gender and race in order to shift visual culture into directions that do not repeat certain undesirable biases from the past. While critiquing the VGAI systems can provide an interesting set of insights, many of the biases have already existed in our visual culture, but are becoming amplified and rendered visible with the use of AI art tools. Therefore, we should use the opportunity of exposing these biases and addressing them not only in the design of the systems but also in a wider and more comprehensive way in our visual culture.

Based on our empirical insights, we outline a few principles that can potentially guide designers and developers to address some of the biases that are clearly visible in the systems. These strategies boil down to a few points, which can be addressed both in training of the AI systems and in the design of the interfaces and interactions with them. The first one being 1) *avoiding defining gender or race based on*

visual characteristics without asking from the subject of representation how they would like to be defined. As we discussed earlier, many of the problems connect to how the interface is labeling ethnicity and gender based on assumptions made in the design process, and producing ‘norms’ through this labeling. To shift away from these practices, designers of these systems can give agency of labeling to the users - albeit this can still also result in undesirable imagery outcomes. Mainly, this practice could aid in shifting away from participating in gendering and racialization of ‘others’. The second suggestion is 2) *decoupling visual features from gender and ethnicity, and presenting them as individual visual features*. In this way, the designers do not engage in automatically coupling visual features with gender or ethnicity, but rather give the users agency in creating representations. These design strategies are commonly used in computer games, in which the users are able to choose various visual features without certain labels being attached to them. This shifts the socio-material arrangement so, that users have more agency to avoid gendering and racialization in creation of content, while arguably they still can engage in producing such imagery by choice. Thirdly, it is important to reflect on making aesthetic judgments in the training of the systems. Our last suggestion is 3) *avoiding coupling aesthetics judgments (good, bad, beautiful, ugly) with certain arrangements of visual features*. As we saw in the radical examples in our paper, attaching such labels to certain gendered or racialized representations effectively produce valuation or devaluation of certain social groups. We hope for these principles, and our study, to spark for further discussion in the HCI community about how we can address visual politics of portrayal in the context of VGAI systems.

6 CONCLUSION

To conclude, these takeaways can serve as a starting point for reflecting on and eventually designing VGAI systems that are sensitive to the visual politics of portrayal. Further careful reflection should be directed toward these systems from the perspective of constructing visual culture and power relations between social categories, and the HCI community is in a central role in developing knowledge and methods for addressing these challenges – and bridging the space between HCI and critical and feminist visual culture knowledge for more equitable futures.

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REFERENCES

- [1] Lisa Adkins and Celia Lury. 1999. The labour of identity: performing identities, performing economies. *Economy and society* 28 (1999), 598–614.
- [2] Taran Anand, Aadi Chauhan, Tanisha Jauhari, Arjav Shah, Rudransh Singh, Benjamin Liang, and Rupsha Dutta. 2023. Identifying race and gender bias in latent diffusion AI image generation. *SSRN Electron. J.* (2023).
- [3] Shaowen Bardzell and Elizabeth F. Churchill. 2011. Editorial: IwC Special Issue "Feminism and HCI: New Perspectives" Special Issue Editors' Introduction. *Interact. Comput.* 23, 5 (sep 2011), iii–xi. [https://doi.org/10.1016/S0953-5438\(11\)00089-0](https://doi.org/10.1016/S0953-5438(11)00089-0)
- [4] Lisa Bloom (Ed.). 1999. *With other eyes*. University of Minnesota Press, Minneapolis, MN. 1–280 pages.
- [5] Alexis I. Boylan. 2020. *Visual culture*. MIT Press, London, England. 1–248 pages.
- [6] Inioluwa Deborah Raji & Joy Buolamwini. 2019. Actionable Auditing: Investigating the Impact of Publicly Naming Biased Performance Results of Commercial AI Products. In *Conference on Artificial Intelligence, Ethics, and Society*.
- [7] Joy Buolamwini. 2023. *Unmasking AI*. Random House, New York, NY. 1–336 pages.
- [8] Lisa Cartwright. 1995. *Screening the body*. University of Minnesota Press, Minneapolis, MN. 1–224 pages.
- [9] Silvia Casale and Vanessa Banchi. 2020. Narcissism and problematic social media use: A systematic literature review. *Addictive Behaviors Reports, Volume 11, 2020, 100252* (2020), 299–305. <https://doi.org/10.1016/j.abrep.2020>
- [10] Mark Coeckelbergh. 2022. *The Political Philosophy of AI – An Introduction*. Polity Press, Cambridge, UK. 1–176 pages.
- [11] Sasha Costanza-Chock. 2020. *Design justice*. MIT Press, London, England.
- [12] Kimberle Crenshaw. 1991. Mapping the Margins: Intersectionality, Identity Politics, and Violence Against Women of Color. *Source: Stanford Law Review* 43, 6 (1991), 1241–1299.
- [13] Andreea Danieleescu. 2020. Eschewing Gender Stereotypes in Voice Assistants to Promote Inclusion. In *Proceedings of the 2nd Conference on Conversational User Interfaces* (Bilbao, Spain) (CUI '20). Association for Computing Machinery, New York, NY, USA, Article 46, 3 pages. <https://doi.org/10.1145/3405755.3406151>
- [14] Susan M. Dray, Daniela K. Busse, Anke Marei Brock, Anicia N. Peters, Shaowen Bardzell, Allison Druin, Margaret M. Burnett, Elizabeth F. Churchill, Gayna Williams, Karen Holtzblatt, and Diane Murray. 2014. Perspectives on Gender and Product Design. In *CHI '14 Extended Abstracts on Human Factors in Computing Systems* (Toronto, Ontario, Canada) (CHI EA '14). Association for Computing Machinery, New York, NY, USA, 53–56. <https://doi.org/10.1145/2559206.2559218>
- [15] Jessica Evans and Stuart Hall. 1999. *Visual culture: The reader*. Sage Publications, Thousand Oaks, CA. 1–512 pages.
- [16] Tony Fry. 2010. *Design as Politics*. Berg, London, England. 1–288 pages.
- [17] Rose Gillian. 2012. *The question of method: practice, reflexivity and critique in visual culture studies*. In: Heywood, Ian and Sandywell, Barry eds. *The Handbook of Visual Culture*. Berg, London, England. 542–558 pages.
- [18] Stuart Hall. 1997. The spectacle of the other. *Representation: Cultural representations and signifying practices* 7 (1997), 225–276.
- [19] Donna Haraway. 1991. *Simians, cyborgs, and women*. Routledge, London, England.
- [20] Amelia Jones (Ed.). 2010. *The feminism and visual culture reader* (2 ed.). Routledge, London, England. 1–736 pages.
- [21] Petra Jääskeläinen, André Holzapfel, and Cecilia Åsberg. 2022. Exploring More-than-Human Caring in Creative-Ai interactions. (2022).
- [22] Petra Jääskeläinen, Daniel Pargman, and André Holzapfel. 2022. On the Environmental Sustainability of Ai Art(s). In *Eighth Workshop on Computing within Limits 2022*. LIMITS. <https://doi.org/10.21428/bf6fb269.c46375fa>
- [23] Shivani Kapania, Alex S. Taylor, and Ding Wang. 2023. A Hunt for the Snark: Annotator Diversity in Data Practices. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, 1–15. <https://doi.org/10.1145/3544548.3580645>
- [24] Alexandra Sasha Luccioni, Christopher Akiki, Margaret Mitchell, and Yacine Jernite. 2023. Stable Bias: Analyzing Societal Representations in Diffusion Models. arXiv:2303.11408 [cs.CY]
- [25] Celia Lury. 2020. The united colors of diversity: essential and inessential culture. In *The Body*. Routledge, Milton Park, UK, 288–293.
- [26] Anne MacClintock. 1995. *Imperial leather: race, gender, and sexuality in the colonial contest*. Routledge, Milton Park, UK. 1–464 pages.
- [27] Laura Mulvey. 1975. *Visual Pleasure and Narrative Cinema*. Afterall Books. 1–64 pages.
- [28] Fabian Offert and Thao Phan. 2022. A Sign That Spells: DALL-E 2, Invisual Images and The Racial Politics of Feature Space. arXiv:2211.06323 [cs.CY]
- [29] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-Resolution Image Synthesis with Latent Diffusion Models. arXiv.
- [30] Morgan Klaus Scheuerman, Kandrea Wade, Caitlin Lustig, and Jed R. Brubaker. 2020. How We've Taught Algorithms to See Identity: Constructing Race and Gender in Image Databases for Facial Analysis. In *Proceedings of the ACM on Human-Computer Interaction*. 1–35. <https://doi.org/10.1145/3392866>
- [31] Katherine Sender and Peter Decherney. 2016. Stuart Hall lives: cultural studies in an age of digital media. *Critical Studies in Media Communication* 33, 5 (2016), 381–384.
- [32] Jaime Snyder. 2020. Visualizing Personal Rhythms: A Critical Visual Analysis of Mental Health in Flux. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference* (, Eindhoven, Netherlands,) (DIS '20). Association for Computing Machinery, New York, NY, USA, 269–281. <https://doi.org/10.1145/3357236.3395463>
- [33] Robert Sparrow. 2020. Robotics Has a Race Problem. *Science Technology and Human Values* 45, 3 (2020), 538–560. <https://doi.org/10.1177/0162243919862862>

- [34] Marita Sturken and Lisa Cartwright. 2017. . Oxford University Press, Oxford, UK. 1–507 pages. Practices of looking.
- [35] Unknown. 2023. Wikimedia Commons, Nazi Germany NSDAP poster. https://commons.wikimedia.org/wiki/File:Nazi_Germany_NSDAP_poster.jpg Original source kerch.ucoz.ua.
- [36] Gloria Wekker. 2016. *White Innocence: Paradoxes of Colonialism and Race*. Duke University Press, Durham, North Carolina, USA. 1–29 pages.
- [37] Cecilia Åsberg. 2005. Genetic Imaginaries: Between Gender and Genes in the visual cultures of popular/ science. Published PhD-Dissertation. (2005).
- [38] Cecilia Åsberg. 2009. Looking at science, looking at you!: the feminist re-visions of nature (Brain and genes). *Teaching Visual Culture in an Interdisciplinary Classroom : Feminist (Re)Interpretations of the Field* (2009), 95–121.
- [39] Cecilia Åsberg. 2024. Promises of Cyborgs: Feminist Practices of Posthumanities (Against the Nested Crises of the Anthropocene). *NORA: Nordic Journal of feminist And gender research, accepted for publication* (2024).
- [40] Cecilia Åsberg and Nina Lykke. 2010. Feminist technoscience studies. *European Journal of Women's Studies* 17.4 (2010), 299–305.